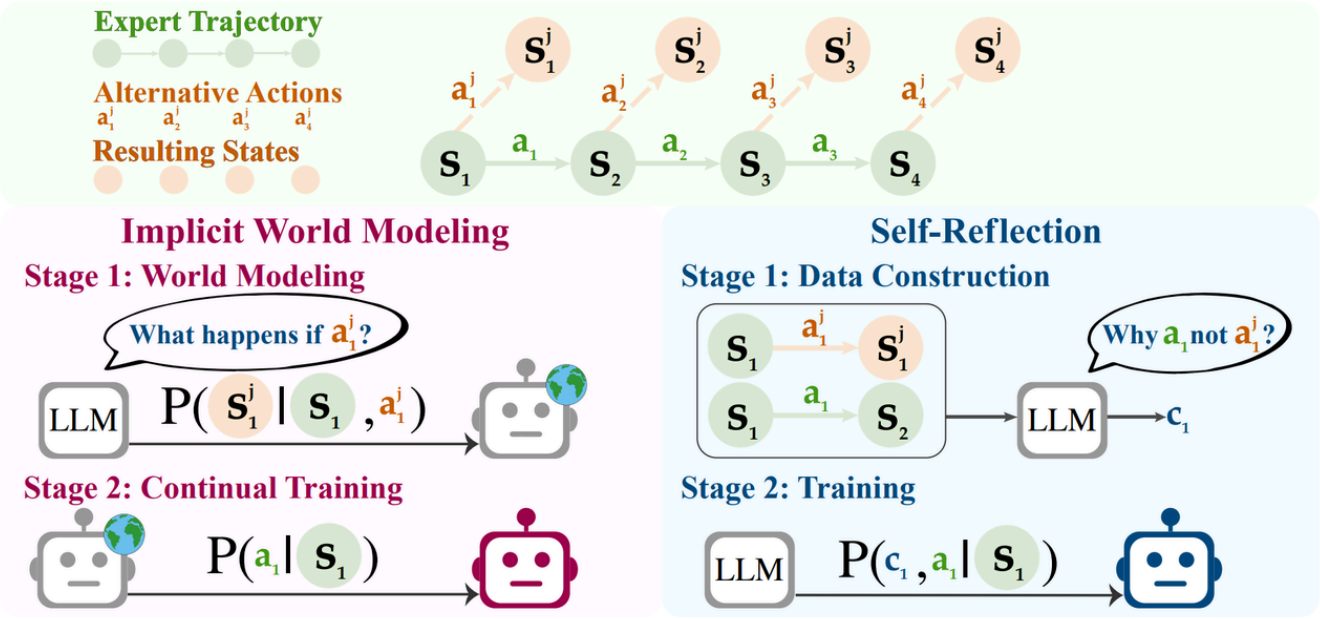
Agent Learning via Early Experience

> #Finetuning #Dataset



- What problems does this framework solve?
  - Currently, finetuning is usually based on humans. That is to say, the dataset and labeling are from humans. It's an imitation, meaning that the model can only learn from what is presented in the dataset, which are the "correct" answers. Thus, during the inference, when there is something wrong, the model will lack the ability to correct.
  - An ideal case is RL. However, for some specific yet complex tasks (such as browser use), there is no effective reward model to provide the training signal.
  - It is observed that the new state after the action is performed can also be a good source of finetuning.
- What does this framework do?
  - This framework provides two paradigms: **Implicit World Modeling** and **Self Reflection**
  - **Implicit world modeling**
    - Use an expert dataset that contains trajectories of actions and states.
    - For each state in the dataset, let our policy model choose its own next actions, a^j.
    - Observe the state change of the environment after performing the action a^j.
    - Make $\{(s_i, a_i^j. s_i^j)\}$ as the dataset.
    - Run SFT with:

    $\mathcal{L}_{IWM} = -\sum_{(s_i, a_i^j, s_i^j) \in \mathcal{D}_{rollout}} \log p_\theta(s_i^j | s_i, a_i^j)$

    - We compare the model's prediction of the result after doing a_i^j to the actual next state s_i^j.\
    - We hope the model learns the dynamics of the environment
  - **Self Reflection**
    - Again, use the expert dataset, but this time, we run the expert action and collect the expert

state change.
    - Still, let the policy model make the decision of the next action and collect the corresponding state change
    - Let a model (or policy model itself) generate a CoT to explain why experts do this and why the world changes in this way based on the difference between the policy model's choice and the expert's choice -> c_i^j
    - Use (s_i, a_i^j, c_i^j) as the dataset to run SFT:
      $\mathcal{L}_{SR} = -\sum \log p_\theta(c_i^j, a_i | s_i)$
    - Here, the label is a_i, the action of the expert, and c_i^j, the CoT. We want the policy model to learn why the expert did it in this way and the reason to do it in this way.
- Experiment and Observations
  1. Comprehensively surpasses imitation learning: Across all testing environments, whether employing “implicit world models” or “self-reflection,” performance consistently outperforms the imitation learning baseline using only expert data. On certain tasks (e.g., WebShop), success rates even increased by 10% to 18%.
  2. Exceptional Data Efficiency: By leveraging early experience methods, performance matching or exceeding conventional imitation learning with full data was achieved using only 1/8th of the expert data volume.
  3. Strong Generalization Capabilities: When confronted with unseen tasks or domains (Out-of-Domain), models trained with early experience demonstrated greater robustness. This stems from learning universal environmental patterns rather than memorizing specific paths.
  4. Excellent stepping stone for reinforcement learning: If the environment eventually provides reward signals (entering the RL phase), using an early experience model as initialization (Warm Start) achieves a higher performance ceiling than starting training from a standard imitation learning model. This demonstrates its effective bridge between imitation learning and reinforcement learning.
- Work Cited
  - Paper
  - Chat History